

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

SEQUENCE DIAGRAMS

UPI Enablement Platform

System Interaction Diagrams Across CBS, PSP Switch, and NPCI

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1. Introduction

This document presents UML-style sequence diagrams for the seven most integration-critical interaction flows of the Deutsche Bank India UPI Enablement Platform. Each diagram shows the participating actors/systems as lifelines and the ordered exchange of messages between them, distinguishing synchronous requests (solid arrows) from responses (dashed arrows). These diagrams are intended for the Integration Team, API Developers, and QA Team to understand call ordering, dependency direction, and timeout/failure handling at a level of detail beyond the BPMN process flows.

1.1 Notation Key

Symbol	Meaning
Vertical dashed line	Lifeline — represents a participant (actor or system) over time
Solid arrow	Synchronous request / call
Dashed arrow	Response / return message
Self-referencing arrow	Internal processing step performed by the same participant (e.g., validation, encryption)

Each diagram is traceable to its corresponding Use Case(s) and Functional Requirement IDs (FR-xx) defined in the Use Case Catalogue and Functional Requirements Specification (FRS), and to the BPMN process of the same business flow in the companion BPMN Process Flows deliverable.

2. UPI Registration

Shows the full registration call sequence from OTP dispatch through consent capture, account discovery against CBS/CIF, and UPI profile creation via the PSP/NPCI layer, concluding with a confirmation notification.

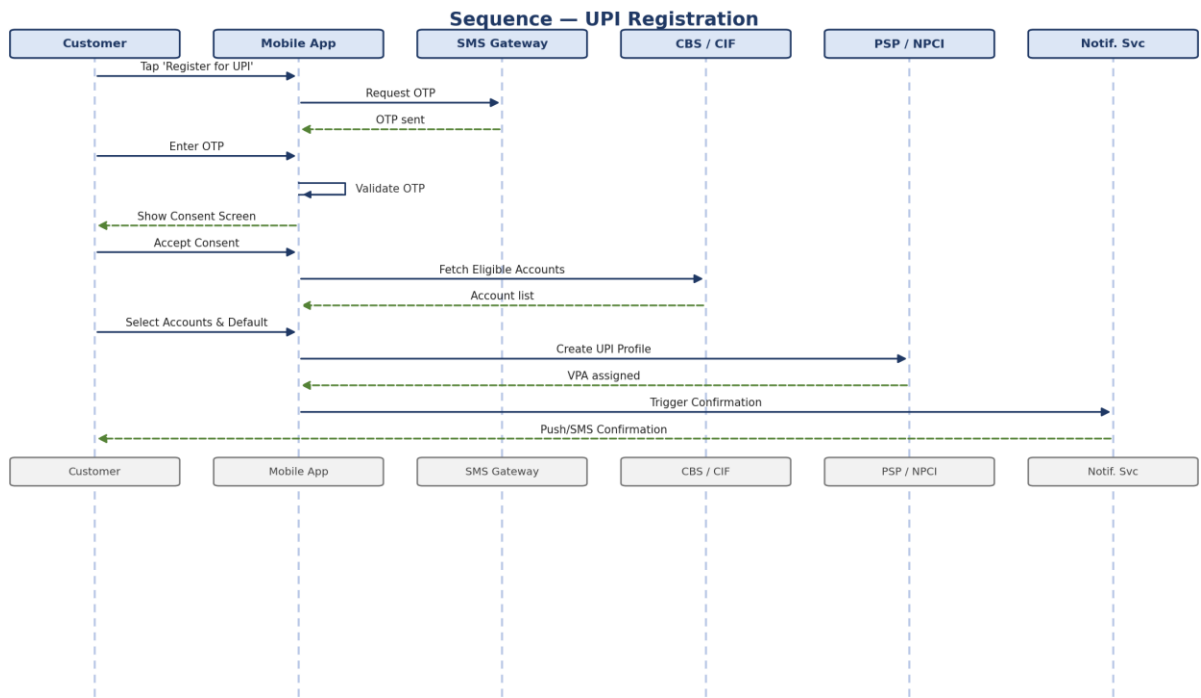


Figure 1 — Sequence: UPI Registration

Related Use Cases / Requirements: UC-01; FR-REG-01 to FR-REG-09

3. UPI PIN Setup

Shows debit-card validation against CBS/CIF, OTP dispatch and verification, on-device PIN encryption, and the PIN-set call to the PSP/NPCI layer.

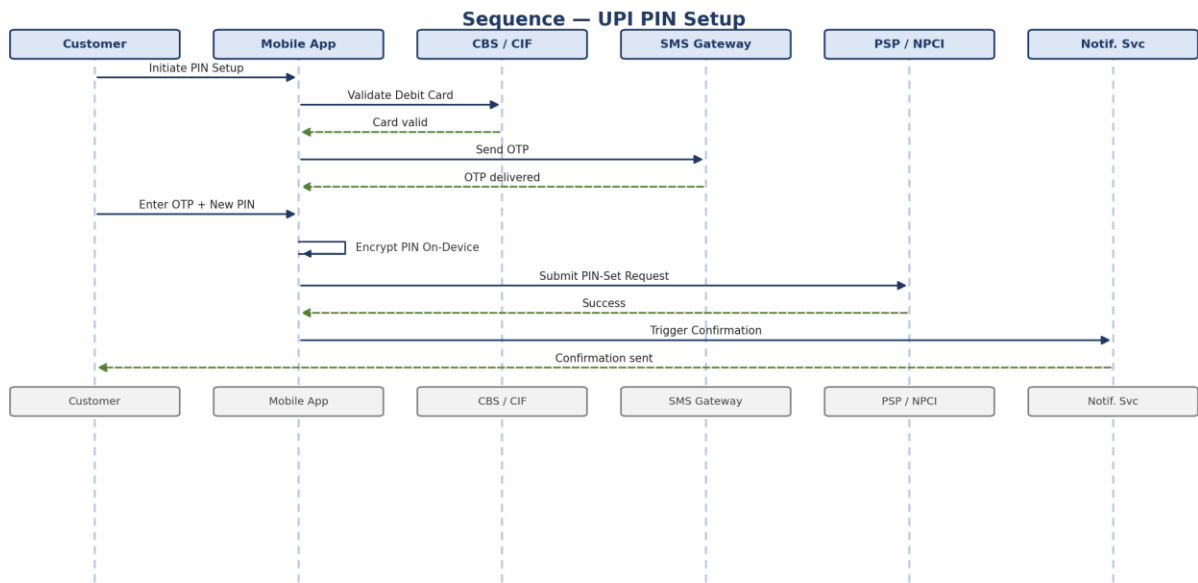


Figure 2 — Sequence: UPI PIN Setup

Related Use Cases / Requirements: UC-06; FR-PIN-01 to FR-PIN-04

4. Send Money — Success Path

Shows the complete successful payment call chain: PIN authorisation, real-time fraud/risk scoring at the PSP switch, NPCI routing, CBS debit posting, and the confirmation/notification return path back to the customer.

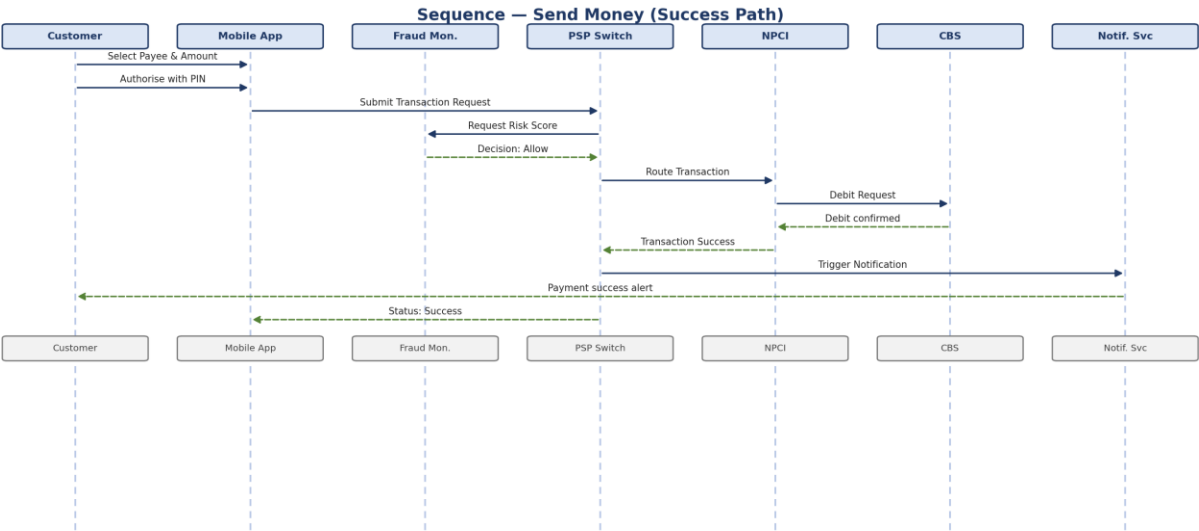


Figure 3 — Sequence: Send Money (Success Path)

Related Use Cases / Requirements: UC-10, UC-11; FR-TXN-06 to FR-TXN-11

5. Send Money — Timeout and Auto-Reconciliation

Shows the failure-handling path where NPCI does not respond within the SLA window: the transaction is marked Pending, a scheduled Deferred Status Check polls for resolution, and an automatic reversal is triggered if a debit occurred without a corresponding credit confirmation.

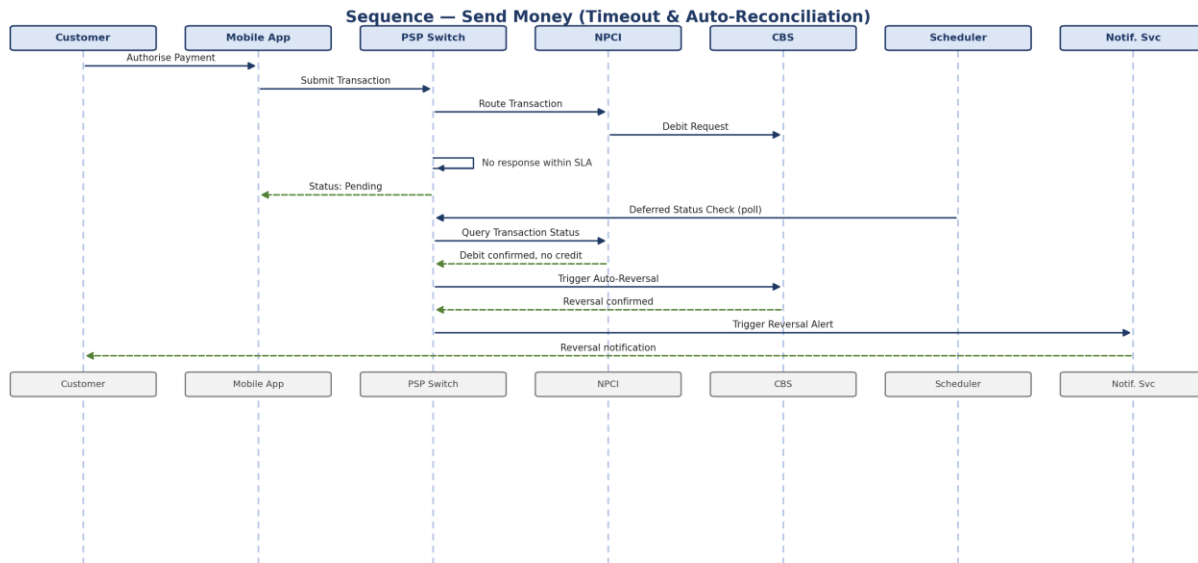


Figure 4 — Sequence: Send Money (Timeout & Auto-Reconciliation)

Related Use Cases / Requirements: UC-14; FR-TXN-14; BR-TXN-08

6. Collect Request — Create and Approve

Shows the collect-request flow from creation and real-time payer notification through payer approval, which internally reuses the standard payment processing sequence.

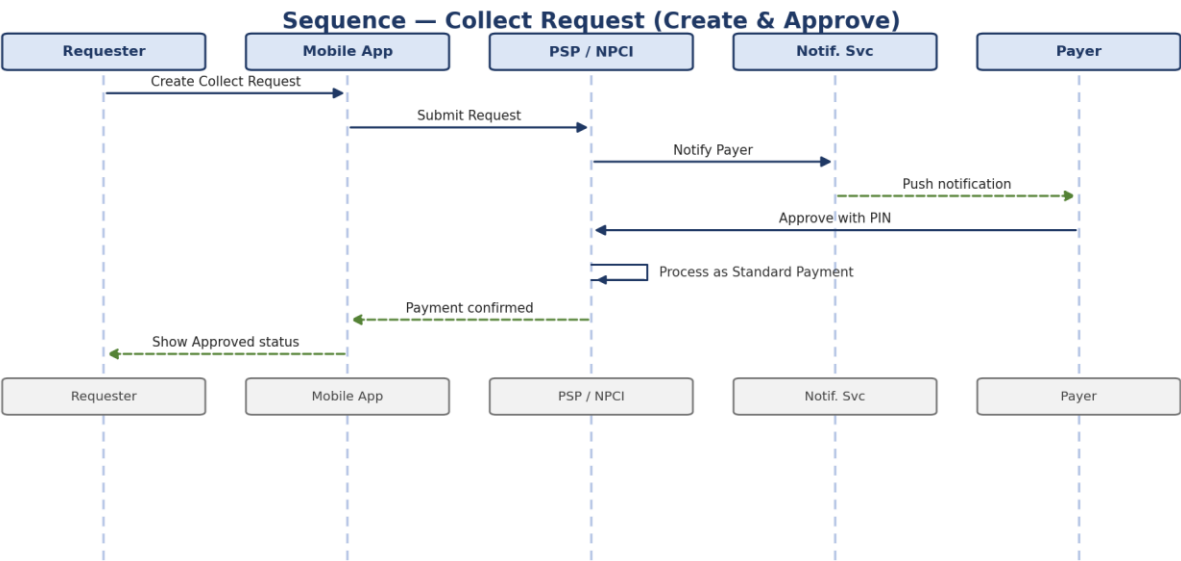


Figure 5 — Sequence: Collect Request (Create & Approve)

Related Use Cases / Requirements: UC-15, UC-16; FR-COL-01 to FR-COL-03

7. Scan & Pay (QR)

Shows QR decoding, merchant VPA validation against the NPCI merchant registry, and the subsequent standard transaction processing sequence through to confirmation.

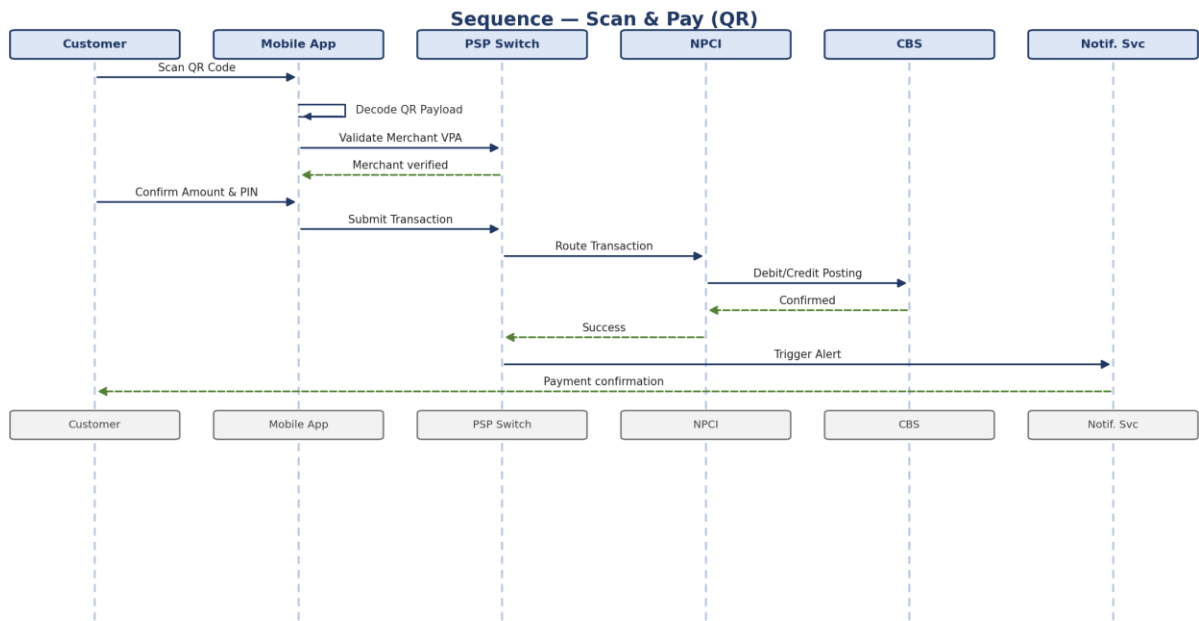


Figure 6 — Sequence: Scan & Pay (QR)

Related Use Cases / Requirements: UC-20; FR-QR-03, FR-QR-04

8. Raise and Resolve Dispute

Shows dispute intake and internal reference generation, conditional escalation to NPCI, Operations investigation, refund initiation via CBS where upheld, and the final customer notification of outcome.

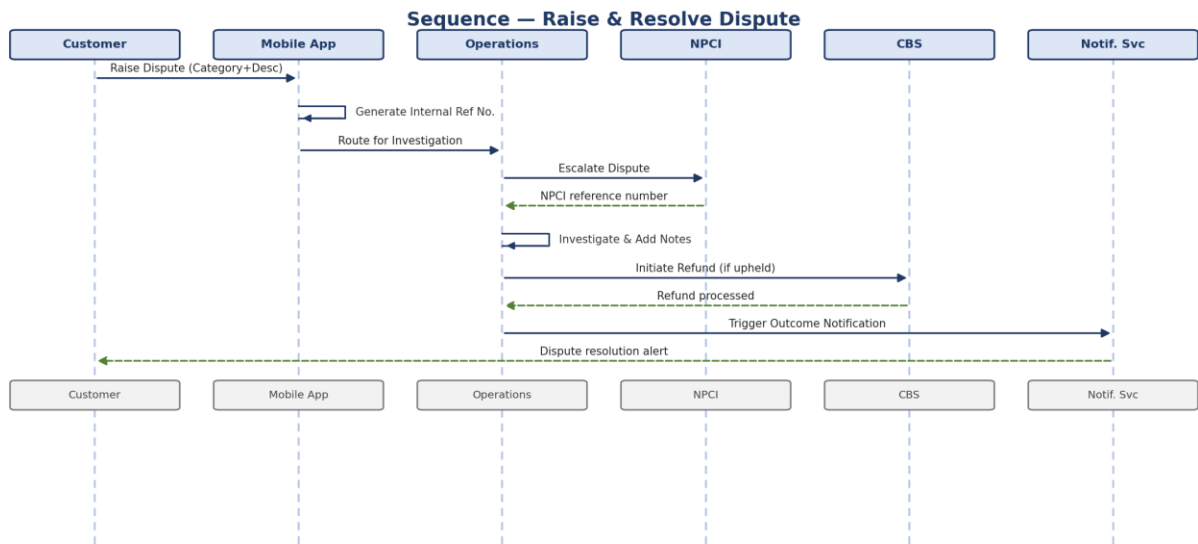


Figure 7 — Sequence: Raise & Resolve Dispute

Related Use Cases / Requirements: UC-26, UC-28; FR-DIS-01 to FR-DIS-06